

Executive Report

Production-Oriented AI Voice Agent for Real Estate

1. Overview

This project is a real, working AI voice agent for real estate — not just a text chatbot. It talks to customers in a natural UrduLish style, answers property questions using real data, gives recommendations, books appointments, and connects with calendar, email, and workflow automation tools.

2. What Was Built

- AI voice agent with UrduLish, Pakistani-style conversation
- Fish Audio text-to-speech for real voice responses
- RAG system using real scraped property data from Zameen.com
- Property database: 200 listings (50 each from Faisalabad, Lahore, Islamabad, Karachi)
- Property recommendation logic based on city, budget, area, bedrooms, amenities
- Conversation memory for customer preferences
- LangGraph workflow to manage the full conversation and tools
- Google Calendar integration for booking, rescheduling, cancellation
- Automated email notifications to real-estate staff
- n8n workflow automation connected via webhook
- Security guardrails against prompt injection and misuse
- FastAPI backend with health and status endpoints
- Monitoring, logging, and deployment setup (Docker, CI, .env)

3. How It Works

Customer speaks or types a request. The AI agent understands the intent, runs a security check, retrieves relevant property data, gives a recommendation or answer, and replies in voice. In the background it can also book a calendar appointment, send an email, or trigger an n8n workflow.

4. Testing & Results

The system was tested using 42 scenarios covering buyers, sellers, investors, appointments, cancellations, rescheduling, off-topic questions, prompt injection, and difficult customers. Key results:

Metric	Result
Evaluation scenarios executed	42
Execution failures	0
Execution success rate	100%
RAG grounding score	20/20 (100%)
Memory accuracy	4/4 (100%)
Fish Audio (voice)	Working
Google Calendar	Working

Email automation	Working
n8n workflow	Working
Average local execution latency	0.0107 seconds

Note: the 0.0107 second figure is local workflow/tool execution time, not full end-to-end voice response time, which depends on network and model speed.

5. Appointment Management

Google Calendar integration supports booking, rescheduling, cancellation, and availability checking. A test slot on 25 September 2026 at 4:00 PM was correctly validated, and unavailable slots were correctly identified.

6. Security

A guardrail built into the LangGraph workflow blocks unsafe requests, such as attempts to reveal internal instructions, access private company data, or create fake appointments. Testing confirmed the guardrail correctly detects and blocks these patterns.

7. Current Status

All core components — conversation, voice, RAG, recommendations, memory, LangGraph workflow, guardrails, Calendar, email, and n8n — have been implemented and tested successfully. The project also includes deployment setup (FastAPI, Dockerfile, CI workflow, environment configuration) so it is ready to move toward full production deployment.

8. Conclusion

This project delivers a real, tested AI voice agent for real estate rather than a simple prototype chatbot. It combines natural voice conversation, grounded property data, recommendations, appointment management, automation, and security into one working system, providing a strong foundation for a full commercial real-estate assistant.